

Exploratory Data Analysis and Visualization of Apparel & Textiles Dataset

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Week: Week 2

Domain: Data Analytics

Dataset: Dress Recommendation Dataset
(OpenML)

Submission Date: (Enter today's date)

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1. Introduction

Exploratory Data Analysis (EDA) is the process of examining and summarizing datasets to understand their characteristics, identify trends, detect anomalies, and prepare the data for further analysis. In the apparel and textiles industry, EDA helps businesses understand consumer preferences, seasonal demand, pricing patterns, and product performance. This report analyzes a publicly available dress recommendation dataset to identify meaningful insights that can support better business decisions.

2. Objective

- Analyze the apparel dataset.
- Identify trends and patterns.
- Visualize important information.
- Generate business insights.

3. Dataset Description

Include:

- **Dataset Name:**
Dress_Recommendation_Cleaned
- **Source:** OpenML
- **File Format:** CSV (cleaned version)
- Number of Rows
- Number of Columns
- Features

Screenshot 1: `df.head()`

	Style	Price	Rating	Size	Season	NeckLine	SleeveLength	WaistLine	Material	FabricType	Decoration	PatternType	Recommendation
0	Sexy	Low	4.6	M	Summer	o-neck	sleeveless	empire	cotton	chiffon	ruffles	animal	2
1	Casual	Low	0.0	L	Summer	o-neck	Petal	natural	microfiber	chiffon	ruffles	animal	1
2	vintage	High	0.0	L	Autumn	o-neck	full	natural	polyester	chiffon	lace	print	1
3	Brief	Average	4.6	L	Spring	o-neck	full	natural	silk	chiffon	embroidary	print	2
4	cute	Low	4.5	M	Summer	o-neck	butterfly	natural	chiffonfabric	chiffon	bow	dot	1

Screenshot 2: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 13 columns):
 #   Column                Non-Null Count  Dtype  
---  --
 0   Style                 500 non-null   object  
 1   Price                 500 non-null   object  
 2   Rating                500 non-null   float64  
 3   Size                  500 non-null   object  
 4   Season                500 non-null   object  
 5   NeckLine              500 non-null   object  
 6   SleeveLength          500 non-null   object  
 7   WaistLine             500 non-null   object  
 8   Material              500 non-null   object  
 9   FabricType            500 non-null   object  
10   Decoration            500 non-null   object  
11   PatternType           500 non-null   object  
12   Recommendation         500 non-null   int64  
dtypes: float64(1), int64(1), object(11)
memory usage: 50.9+ KB
```

4. Exploratory Data Analysis

Describe:

- Dataset shape
- Data types
- Missing values
- Duplicate records
- Summary statistics

Include:

Screenshot 3: `df.describe()`

	Rating	Recommendation
count	500.000000	500.000000
mean	3.528600	1.420000
std	2.005364	0.494053
min	0.000000	1.000000
25%	3.700000	1.000000
50%	4.600000	1.000000
75%	4.800000	2.000000
max	5.000000	2.000000

Screenshot 4: `df.isnull().sum()`

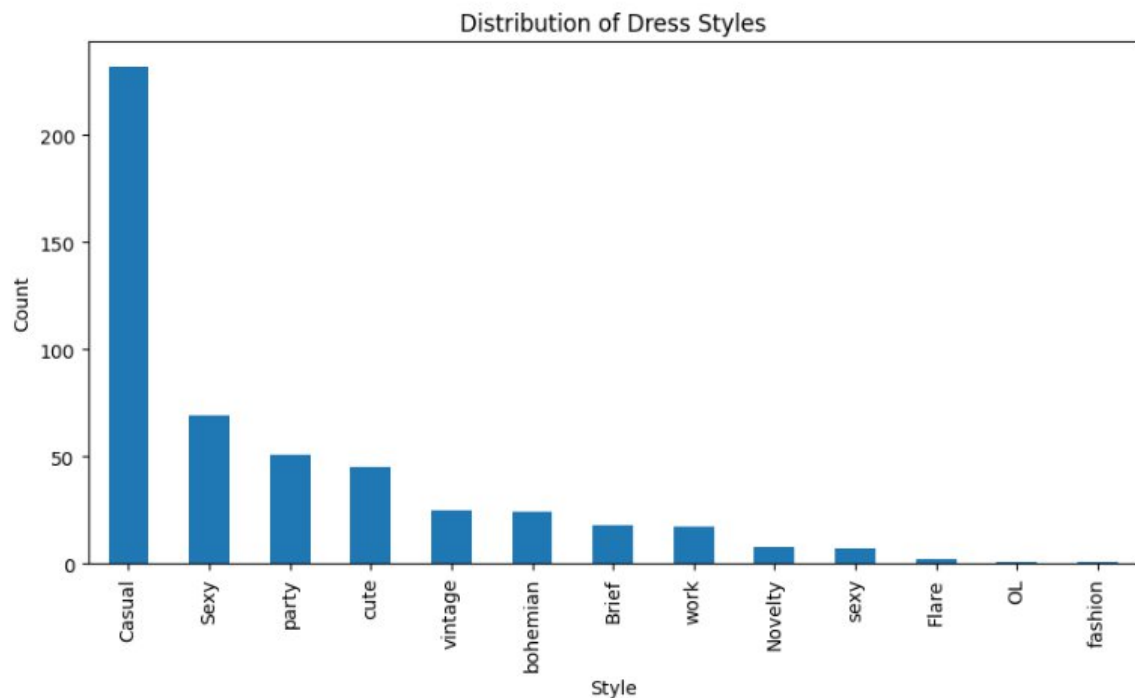
Style	0
Price	0
Rating	0
Size	0
Season	0
NeckLine	0
SleeveLength	0
WaistLine	0
Material	0
FabricType	0
Decoration	0
PatternType	0
Recommendation	0

dtype: int64

5. Data Visualization

Visualization 1 – Bar Chart

Title: Dress Style Distribution



Observation:

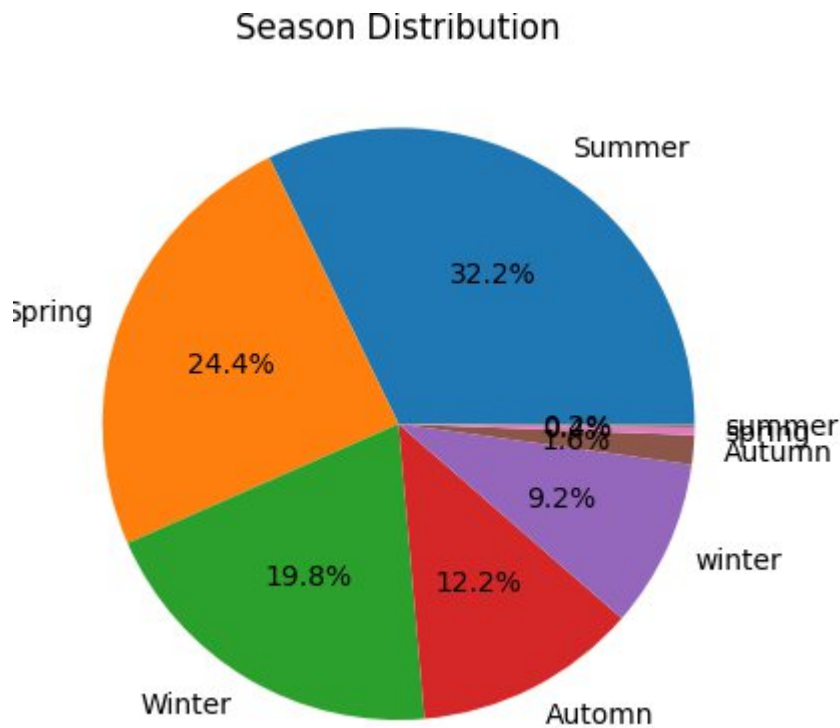
- Casual and Sexy styles appear frequently.

Business Insight:

- These styles may represent popular customer preferences.

Visualization 2 – Pie Chart

Title: Season Distribution



Observation:

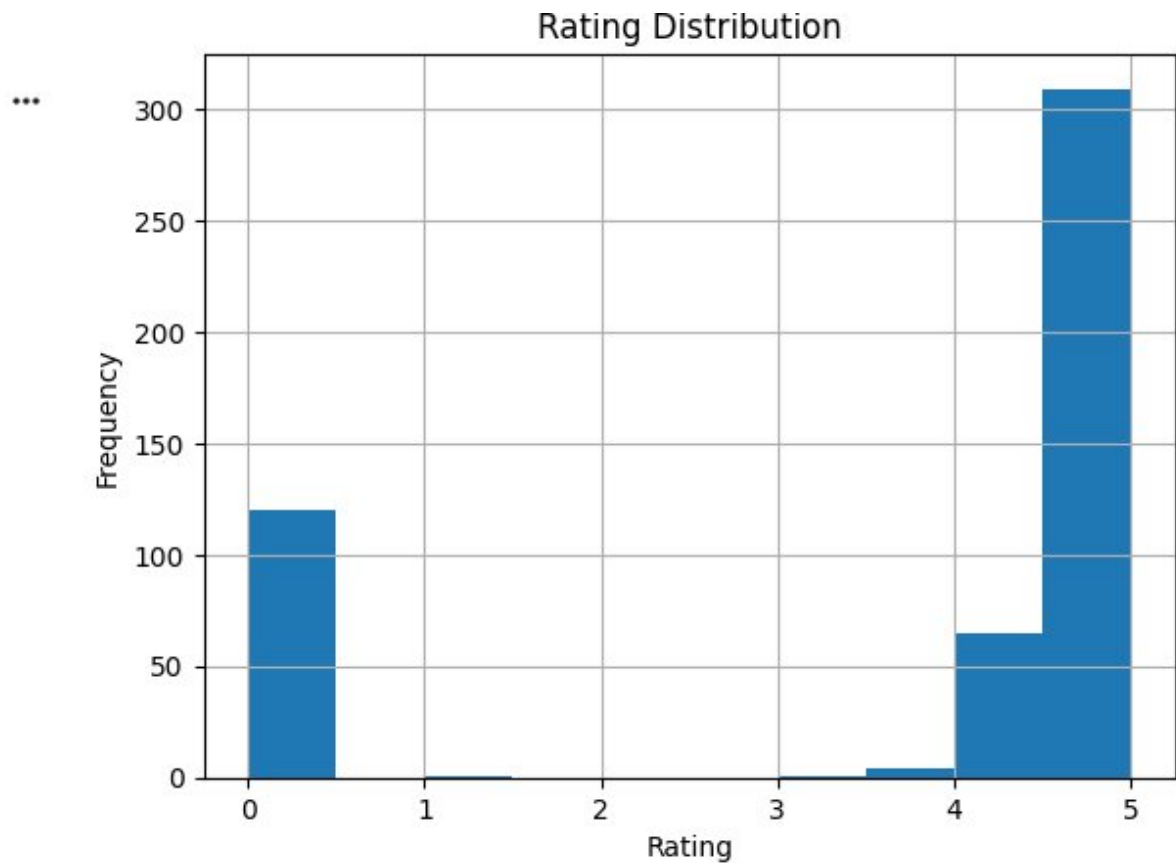
- Summer dresses account for a large portion of the dataset.

Business Insight:

- Retailers should plan inventory before the summer season.

Visualization 3 – Histogram

Title: Rating Distribution



Observation:

- Most products have ratings between 4 and 5.

Business Insight:

- High-rated products should be promoted more aggressively.

6. Key Findings

Write in bullet points.

- Summer dresses dominate the dataset.
- Low-price products are more common.
- Most customer ratings are positive.
- Casual and Sexy styles are popular.

7. Business Implications

- Increase inventory for popular styles.
- Plan seasonal promotions before summer.
- Promote highly rated products.
- Review less popular categories for improvement.

8. Challenges

Mention:

- Missing values
- Byte-encoded text
- ARFF file format

- Categorical data cleaning

9. Limitations

- Public dataset
- No real-time sales information
- Limited geographical coverage
- Some missing attributes

10. Conclusion

Conclusion

This exploratory data analysis (EDA) successfully examined the Dress Recommendation Dataset to identify meaningful patterns, trends, and insights within the apparel and textiles industry. The dataset was analyzed using Python in Google Colab with the help of libraries such as pandas and Matplotlib. Initial data exploration included reviewing the dataset structure, verifying data types, handling missing values,

removing duplicate records, and preparing the data for analysis.

Several visualizations, including bar charts, pie charts, and histograms, were created to better understand the distribution of dress styles, seasonal preferences, price categories, and product ratings. The analysis revealed that certain dress styles and seasons appeared more frequently than others, indicating variations in consumer preferences. The rating distribution also showed that many products received favorable ratings, suggesting generally positive customer feedback.

From a business perspective, these findings can help apparel companies improve inventory planning, optimize seasonal product launches, focus marketing efforts on popular product categories, and promote highly rated items. The analysis also highlighted the importance of maintaining clean and well-structured data, as accurate data preparation is essential for producing reliable insights.

Although the dataset has limitations, such as the absence of real-time sales data and limited market coverage, it provides a strong foundation for understanding consumer behavior and product characteristics. Overall, this project demonstrates how exploratory data analysis can transform raw data into valuable business insights and support informed decision-making in the apparel and textiles industry.

11. References

Include:

- OpenML
- Pandas Documentation
- Matplotlib Documentation
- Google Colab